



Bank Customer Churn Analysis



Introduction

Bank customer churn is a major challenge where customers close their bank accounts, directly impacting the bank's revenue and growth. This project uses the Bank_Churn dataset to analyze customer demographics, account activity, and behavior patterns. The main aim is to identify factors and patterns that trigger customer churn. SQL queries have been used to explore customer segments, exit rates, and key trends within the data. This analysis will help the bank develop better customer retention strategies.

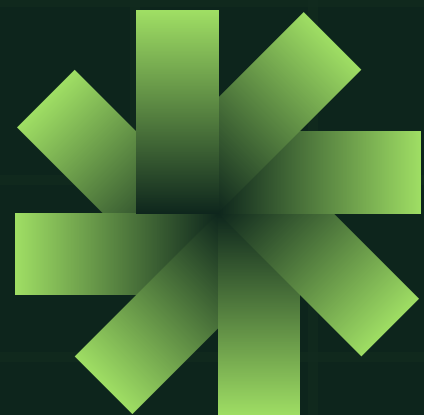
Objective

The objective of this project is to understand customer churn behavior and identify high-risk customer segments. Exit rates have been analyzed based on Age, Gender, and Geography to find patterns. The impact of factors like Credit Score, Balance, and Product usage on churn has also been examined. A comparison between active and inactive members was done to understand engagement patterns. The overall goal is to provide data-driven insights that help improve customer retention.

Dataset Overview

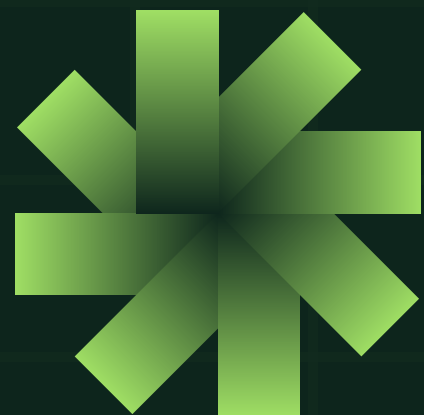
The dataset contains the following columns:

- CustomerId
- Surname
- CreditScore
- Geography
- Gender
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary
- Exited
- Age_Group



Business Problems

- Q1. What is the total number of customers?
- Q2. What is the average credit score of customers?
- Q3. What is the total number of active members?
- Q4. What is the total number of customers who exited?
- Q5. How many customers have a credit card?
- Q6. What is the Average Balance of customers?
- Q7. Which geography has the most customers with a credit card?
- Q8. How many years did customers stay before exiting?
- Q9. Which customers are using more than 2 products?
- Q10. Who are the customers with the highest credit scores?
- Q11. Which geography has the highest average credit score?
- Q12. Which age group has the most customers with a credit card?
- Q13. Which age group has the highest customer exit rate?
- Q14. Which geography has the highest customer exit rate?
- Q15. Which gender has a higher exit rate?
- Q16. What is the number of Active and Exited Customers in each Geography?



SQL Queries & Analysis



Q1. What is the total number of customers?

```
Select  
    COUNT(Distinct CustomerId) as Total_Customers  
From Bank_Churn
```

Results		Messages	
	Total_Customers		
1	10000		

SQL Queries & Analysis

Q2. What is the average credit score of customers?



```
Select  
    AVG(CreditScore) as Avg_Credit_Score  
From Bank_Churn
```



Results



Messages

	Avg_Credit_Score
1	650

SQL Queries & Analysis



Q3. What is the total number of active members?

```
Select  
    COUNT(IsActiveMember) as Total_Active_Members  
From Bank_Churn  
where IsActiveMember = 1
```

Results		Messages	
	Total_Active_Members		
1	5151		

SQL Queries & Analysis

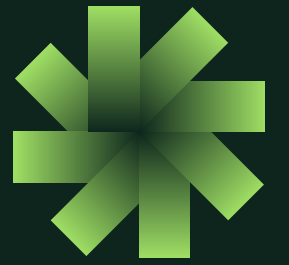
Q4. What is the total number of customers who exited?



```
Select  
    COUNT(Exited) as Total_Exited  
From Bank_Churn  
where Exited = 1
```

Results		Messages	
	Total_Exited		
1	2037		

SQL Queries & Analysis

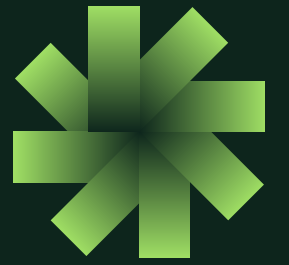


Q5. How many customers have a credit card?

```
Select  
    COUNT(HasCrCard) as Total_Credit_Card  
From Bank_Churn  
where HasCrCard = 1 and IsActiveMember = 1
```

Results		Messages	
	Total_Credit_Card		
1	3607		

SQL Queries & Analysis



Q6. What is the Average Balance of customers?

```
Select  
    Round(AVG(Balance),3) as Avg_Balance  
From Bank_Churn
```

Results		Messages	
	Avg_Balance		
1	76485.889		

SQL Queries & Analysis

Q7. Which geography has the most customers with a credit card?



```
Select
    Geography,
    count(HasCrCard) as Credit_Card
From Bank_Churn
where HasCrCard = 1
Group by Geography
Order by Credit_Card desc
```

Results		Messages
	Geography	Credit_Card
1	France	3543
2	Germany	1791
3	Spain	1721

SQL Queries & Analysis

Q8. How many years did customers stay before exiting?



```
Select  
    CustomerId,  
    Tenure  
From Bank_Churn  
Where Exited = 1  
Group by CustomerId, Tenure
```

Results		Messages
	CustomerId	Tenure
1	15565706	1
2	15566030	5
3	15566211	1
4	15566251	5
5	15566253	9
6	15566312	5
7	15566378	5
8	15566495	2
9	15566531	4
10	15566563	4
11	15566708	4
12	15566988	7

SQL Queries & Analysis

Q9. Which customers are using more than 2 products?



```
Select CustomerId, NumOfProducts From Bank_Churn  
Where NumOfProducts >2;
```

Results		Messages
	CustomerId	NumOfProducts
1	15619304	3
2	15656148	4
3	15589475	3
4	15703793	4
5	15622897	3
6	15757535	3
7	15676895	3
8	15764866	3
9	15794056	3
10	15619955	3
11	15684173	3
12	15583456	3

SQL Queries & Analysis

Q10. Who are the customers with the highest credit scores?



```
SELECT CustomerId, Surname, CreditScore
FROM Bank_Churn
WHERE CreditScore = (SELECT MAX(CreditScore) FROM Bank_Churn);
```

Results		Messages	
	CustomerId	Surname	CreditScore
1	15737888	Mitchell	850
2	15717426	Armstrong	850
3	15716334	Rozier	850
4	15604482	Chiemezie	850
5	15733247	Stevenson	850
6	15607178	Welch	850
7	15611759	Simmons	850
8	15760431	Pino	850
9	15785798	Uchechukwu	850
10	15663252	Olisanugo	850
11	15671256	Macartney	850
12	15653517	Madukwa	850

SQL Queries & Analysis



Q11. Which geography has the average credit score?

```
Select
    Geography,
    AVG(CreditScore) as Avg_Credit_Score
from Bank_Churn
Group by Geography
```

	Geography	Avg_Credit_Score
1	Spain	651
2	France	649
3	Germany	651

SQL Queries & Analysis

Q12. Which age group has the most customers with a credit card?



```
Select
    Age_Group,
    COUNT(HasCrCard) as Credit_Card
from Bank_Churn
Where HasCrCard = 1
Group by Age_Group
Order by Credit_Card desc;
```

Results Messages		
	Age_Group	Credit_Card
1	35 - 45	2812
2	25 - 35	2296
3	45 - 55	1007
4	55 - 65	423
5	18 - 25	316
6	65 - 75	165
7	75 - 85	33
8	85 - 95	3

SQL Queries & Analysis

Q13. Which age group has the highest customer exit rate?



```
With cte as (  
  SELECT  
    Age_Group,  
    SUM(Case When Exited = 1 Then 1 Else 0 End) AS Total_Exited,  
    CAST(  
      100.0 * SUM(Case When Exited = 1 Then 1 Else 0 End)  
      / SUM(SUM(Case When Exited = 1 Then 1 Else 0 End)) OVER ()  
      AS DECIMAL(5,2)  
    ) AS Exit_Rate  
  FROM Bank_Churn  
  GROUP BY Age_Group  
)  
select Age_Group, Exit_Rate from cte  
ORDER BY Exit_Rate DESC;
```

Results Messages		
	Age_Group	Exit_Rate
1	35 - 45	34.51
2	45 - 55	34.46
3	55 - 65	14.68
4	25 - 35	12.27
5	65 - 75	2.06
6	18 - 25	1.96
7	75 - 85	0.05
8	85 - 95	0.00

SQL Queries & Analysis

Q14. Which geography has the highest customer exit rate?



```
SELECT
    Geography,
    Round(CAST(100.0 * SUM(CASE WHEN Exited = 1 THEN 1 ELSE 0 END) As float) / COUNT(*),2) AS Exit_Rate
FROM Bank_Churn
GROUP BY Geography
ORDER BY Exit_Rate DESC;
```

 Results  Messages

	Geography	Exit_Rate
1	Germany	32.44
2	Spain	16.67
3	France	16.15

SQL Queries & Analysis

Q15. Which gender has a higher exit rate?



```
With Cte as (  
  SELECT  
    Gender,  
    SUM(Case When Exited = 1 Then 1 Else 0 End) AS Total_Exited,  
    CAST(  
      100.0 * SUM(Case When Exited = 1 Then 1 Else 0 End)  
      / SUM(SUM(Case When Exited = 1 Then 1 Else 0 End)) OVER ()  
      AS DECIMAL(5,2)  
    ) AS Exit_Rate  
  FROM Bank_Churn  
  GROUP BY Gender  
)  
Select Gender, Exit_Rate from Cte  
ORDER BY Exit_Rate DESC;
```

Results		Messages
	Gender	Exit_Rate
1	Female	55.92
2	Male	44.08

SQL Queries & Analysis



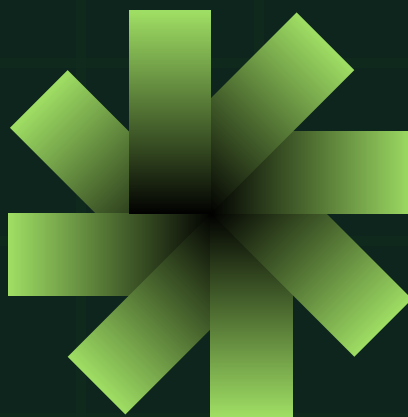
Q16. What is the number of Active and Exited Customers in each Geography?

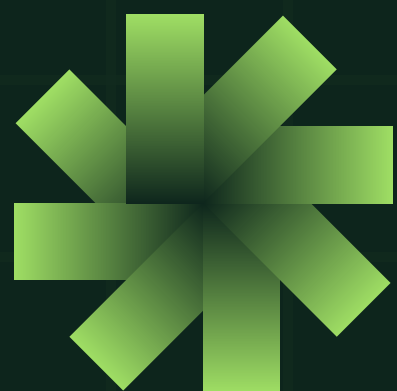
```
Select
    Geography,
    SUM(Case When IsActiveMember = 1 Then 1 Else 0 End) as Active_Customers,
    SUM(Case When Exited = 1 Then 1 Else 0 End) as Exited_Customers
from Bank_Churn
Group by Geography
```

Results		Messages	
	Geography	Active_Customers	Exited_Customers
1	Spain	1312	413
2	France	2591	810
3	Germany	1248	814

Insights

- There are 10,000 total customers, of which 5,151 are active (51.5%) and 2,037 have exited (20.4%).
- France has the highest number of customers using credit cards, with 3,543 customers, followed by Germany with 1,791 and Spain with 1,721.
- The majority of exits occurred among customers aged 35–45 (34.51%), followed closely by the 45–55 age group (34.46%).
- The highest credit card usage is among customers aged 35–45, with 2,812 customers, followed by the 25–35 age group, with 2,296 customers.
- Only 326 customers are using more than two products across the bank's services.
- Female customers account for a higher share of total churn (55.92%) compared to male customers (44.08%).
- Exited customers stayed with the bank for an average of 4.9 years before churning.





Thank You

